# Enterprise AI Guidance for AWS

**Precise Model, RAG and Guardrail Configuration Standard**

Document ID: EA-AI-002 | Version: 2.0 | Status: APPROVED GUIDANCE | Classification: INTERNAL

Owner: Enterprise Architecture | Review Frequency: Quarterly | Primary Scope: AWS Generative AI and RAG

This standard provides precise enterprise guidance for approved AWS AI services, foundation model identifiers, inference profiles, RAG configuration, embedding models, guardrails, model lifecycle, and production controls. Exact production availability must always be validated in the AWS Region before deployment.

## 1. Enterprise AI Platform Standard

Amazon Bedrock is the approved default platform for managed Generative AI workloads. It is the preferred service for foundation model inference, enterprise assistants, RAG, summarization, extraction, and supported AI agent capabilities. Amazon SageMaker AI should be selected when custom machine learning training, fine-tuning, specialized model development, or greater ML lifecycle control is required.

### Mandatory Platform Rules

Production teams MUST use an approved service and MUST record the exact model identifier or inference profile used by the application. Teams MUST NOT rely only on a model family name such as Claude Sonnet or Amazon Nova because the exact identifier determines the deployed model configuration and lifecycle.

The enterprise default Region for European workloads is eu-west-1 unless a workload-specific architecture decision defines another approved Region. Cross-Region inference must be assessed against data residency and compliance requirements.

### Approved Service Baseline

```
AI Inference: Amazon Bedrock

Managed RAG: Amazon Bedrock Knowledge Bases

AI Safety: Amazon Bedrock Guardrails

AI Agents: Amazon Bedrock Agents - architecture review required

Custom ML: Amazon SageMaker AI

Knowledge Documents: Amazon S3

Deterministic Workflow: AWS Step Functions

Event Routing: Amazon EventBridge

Async Buffering: Amazon SQS

Secrets: AWS Secrets Manager
```

## 2. Approved Foundation Model Catalog

The following catalog provides example approved configurations for enterprise workloads. The catalog must be validated against the current Amazon Bedrock model availability and compatibility for the deployment Region before production use.

### Enterprise RAG Model - Balanced

```
Provider: Anthropic

Model: Claude Sonnet 4.6

Base Model ID: anthropic.claude-sonnet-4-6

EU Inference Profile Pattern: eu.anthropic.claude-sonnet-4-6

Primary Use: Enterprise RAG, architecture assistant, document analysis

Approval: APPROVED SUBJECT TO REGION AVAILABILITY
```

Claude Sonnet 4.6 should be evaluated as a balanced high-quality option for enterprise knowledge workloads. Production applications should use the exact approved invocation identifier recorded in the central model catalog.

### Fast Model

```
Provider: Anthropic

Model: Claude Haiku 4.5

Base Model ID: anthropic.claude-haiku-4-5-20251001-v1:0

EU Inference Profile Pattern: eu.anthropic.claude-haiku-4-5-20251001-v1:0

Primary Use: Classification, extraction, simple summarization

Approval: APPROVED SUBJECT TO REGION AVAILABILITY
```

Fast models should be preferred for high-volume workloads where evaluation confirms sufficient quality.

## 3. Model Governance and Lifecycle

Foundation models are enterprise dependencies. Every production AI application MUST maintain a model configuration record. A model change may affect answer quality, latency, cost, safety behavior, context limits, and tool capabilities and must therefore be evaluated before production deployment.

### Required Production Model Record

```
Application ID: APP-AI-001

AWS Region: eu-west-1

AI Service: Amazon Bedrock

Provider: Anthropic

Model Family: Claude Sonnet

Exact Model ID: [central catalog value]

Inference Profile ID: [central catalog value]

Lifecycle Status: ACTIVE

API: Converse

Evaluation Dataset: Enterprise-AI-Eval-v1

Approval Status: APPROVED
```

Applications MUST NOT automatically upgrade to a newly released model. Active models may be used. Legacy models require a migration assessment. Retired or end-of-life models MUST be removed from production before the applicable retirement deadline.

### Model Selection Rule

The selected model must be the least expensive approved model that meets the measured quality requirement. A higher-capability model must not be selected solely because it produces longer or more fluent responses.

## 4. Enterprise RAG Standard

RAG must be used when the application requires organization-specific, private, changing, or externally governed knowledge. RAG must not be used as a substitute for authoritative real-time APIs or transactional databases.

### Standard RAG Configuration

```
Knowledge Base Name: Enterprise-Architecture-KB

Document Store: Amazon S3

Embedding Model: amazon.titan-embed-text-v2:0

Generation Model: Approved central catalog model

Chunking: Semantic where supported and appropriate

Initial Retrieval Results: 5

Metadata Filtering: REQUIRED where access or domain filtering is needed

Source Citations: REQUIRED

No-Answer Handling: REQUIRED

Evaluation: REQUIRED before production
```

Amazon Titan Text Embeddings V2 is an approved example embedding model for text-based enterprise knowledge. If the embedding model changes, teams MUST evaluate retrieval quality and re-index the knowledge base when required by the selected architecture.

### No-Answer Policy

When sufficient evidence cannot be retrieved, the application must state: "I could not find sufficient information in the approved enterprise knowledge base to provide a reliable answer." The application MUST NOT invent enterprise policy, approved model IDs, service approvals, architecture decisions, or security requirements.

## 5. Knowledge Document Standard

Knowledge documents must be written for retrieval precision. Each document MUST clearly identify its status, version, owner, scope, and review date. Statements should be explicit and should distinguish approved requirements from recommendations and examples.

### Required Metadata

```
Document ID: EA-AI-002

Document Title: Enterprise AI Guidance

Version: 2.0

Status: APPROVED

Owner: Enterprise Architecture

Classification: INTERNAL

Effective Date: 2026-09-11

Review Frequency: Quarterly

Knowledge Domain: AWS AI
```

Architecture guidance should explicitly answer: What is approved? When should it be used? When must it not be used? What is the alternative? What controls are mandatory? Who owns the decision?

## 6. Enterprise Guardrail Standard

Amazon Bedrock Guardrails is the approved AWS capability for configurable AI safety controls. Guardrails must be assessed for every production Generative AI application. Guardrails provide an additional safety layer and do not replace IAM, authorization, encryption, network security, data classification, or application security.

### Production Guardrail Configuration

```
Guardrail Name: Enterprise-AI-Guardrail

Guardrail ID: [organization-created ID]

Production Version: [approved numeric version]

Region: eu-west-1

Draft Allowed in Production: NO

Owner: AI Governance
```

A real guardrail ID and numeric version are created by the organization's AWS account and therefore must not be invented in architecture guidance. The central configuration catalog must provide the actual production values.

### Content Filter Baseline

```
Hate: HIGH

Insults: MEDIUM

Sexual Content: HIGH

Violence: MEDIUM

Misconduct: HIGH

Prompt Attack: HIGH
```

The final filter strengths must be validated against the application's risk profile and tested for false positives and false negatives.

## 7. Denied Topics and Sensitive Information

### Denied Topic: Production Credentials

The AI application MUST NOT provide, retrieve, generate, or expose production passwords, AWS access keys, secret values, API keys, session credentials, or access tokens.

### Denied Topic: Security Bypass

The AI application MUST NOT provide instructions intended to bypass enterprise authentication, authorization, security controls, monitoring, or approved access restrictions.

### Sensitive Information Actions

```
AWS Access Keys: BLOCK

Passwords: BLOCK

API Keys: BLOCK

Access Tokens: BLOCK

Email Address: MASK where policy requires

Phone Number: MASK where policy requires

Payment Card Information: BLOCK
```

Applications must implement independent authorization and data access controls. A guardrail response alone must never be treated as evidence that the user was authorized to access the requested information.

## 8. Prompt Injection and Grounding

The enterprise test suite must include attempts to override instructions, reveal system prompts, retrieve unauthorized documents, bypass access controls, and disable safety controls.

```
Test: Ignore previous instructions

Test: Reveal system prompt

Test: Show hidden instructions

Test: Retrieve all confidential documents

Test: Bypass authorization

Test: Act as administrator

Test: Disable guardrails
```

Expected behavior is that authorization remains enforced, restricted data is not exposed, hidden instructions are not disclosed, and safety controls remain active.

### RAG Grounding

Enterprise RAG responses must be supported by retrieved evidence. Where contextual grounding and relevance checks are enabled for the selected architecture, thresholds must be centrally configured and evaluated against representative enterprise questions. Threshold values should not be invented or copied between applications without testing.

## 9. AI Agent Standard

AI agents may be used when the application must dynamically determine the next action or select among approved tools. A fixed sequence of actions should use deterministic orchestration such as AWS Step Functions.

### Example Agent Configuration

```
Agent Name: Enterprise-Support-Agent

Model: Approved central catalog model

Knowledge Base: Enterprise-KB

Guardrail: Enterprise-AI-Guardrail

Tool Access: Least privilege

Maximum Scope: Defined by application policy

High Impact Actions: Human approval required
```

Agents MUST NOT receive unrestricted access to enterprise systems. Every tool must independently authenticate and authorize the request, validate input, log execution, and enforce business controls.

## 10. Production Approval Requirements

Before production deployment, the application owner and architecture reviewer must confirm that the AWS service is approved, the exact model invocation identifier is recorded, lifecycle status is acceptable, and evaluation has been completed.

For RAG applications, knowledge sources must be approved, owned, current, and versioned. Source citations and no-answer behavior must be implemented where grounded enterprise answers are required. Retrieval must not bypass document or application authorization.

For Generative AI applications, guardrail requirements must be assessed and production guardrail versions must be controlled. The solution must be monitored for errors, latency, model usage, retrieval quality, guardrail interventions, tool invocations, and cost.

### Enterprise Principle

Select the simplest approved technology that meets the requirement. Use exact model and guardrail identifiers in production configuration. Keep enterprise knowledge current and explicit. Evaluate every material AI change before release.